

Step-by-Step Solutions: Unit Test 01 Hydrocarbons

Official Verified Marking Scheme by Ajay Choudhary Sir

Chapter / Focus Area: Organic Chemistry - Model Solutions & Rubrics

Detailed Solutions & Marking Rubrics

Ans 1. (a) $(\text{CH}_3)_3\text{C}^+$ - 3° Carbocation with 9 hyperconjugative α -hydrogens. [1 Mark]

Ans 2. (b) Ethanal (CH_3CHO) - Symmetrical cleavage at double bond. [1 Mark]

Ans 3. Mechanism: Electrophilic addition of H^+ creates 2° carbocation intermediate ($\text{CH}_3\text{-CH}^+\text{-CH}_3$), followed by fast nucleophilic attack of Br^- . [2 Marks]

Ans 4. Step 1: Benzene + Conc. HNO_3 + Conc. H_2SO_4 at 330 K $\rightarrow$ Nitrobenzene. Step 2: Nitrobenzene + Sn/HCl (or Fe/HCl) $\rightarrow$ Aniline. [3 Marks]

Ans 6. Compound A is 1-Bromo-2-methylpropane / tert-Butyl bromide. Compound B is 2-Methylpropene ($\text{CH}_3)_2\text{C=CH}_2$. Ozonolysis cleaves C=C giving acetone and formaldehyde. [5 Marks]

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